Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*			${}^{228}_{90}\mathrm{Th} \rightarrow {}^{224}_{88}\mathrm{Ra} + {}^{4}_{2}\alpha$ (1) ${}^{90}_{38}\mathrm{Co} \rightarrow {}^{90}_{39}\mathrm{Y} + {}^{0}_{-1}\beta$ (1)		2		2		
	(b)			Nucleon mass = 90.727 u **or** nucleon mass +38 e (90.747u) (1) Mass defect attempted with or without electrons 0.84 u or 0.82 u (1) × 931 and division by 90 (1) Answer = 8.69 [MeV per nucleon] (1) If electrons not taken into account answer = 8.47 [MeV per nucleon] award 3 marks 782 or 762 [MeV per nucleon] award 2 marks	1	1 1 1		4	3	
	(c)	(i)		Probability of landing on black face = ¼ **or** 0.25 **or** 25 %		1		1	1	1
		(ii)	I.	Probability of not decaying (i.e. of remaining) after 1 throw = 1 – 0.25 = 0.75 (1) Probability of remaining after 2 throws = 0.75^2 **or** probability of remaining after n throws = $(0.75)^n$ (1)		2		2	2	2
			II.	Number predicted = $N_0 \times (0.75)^n$ = 31.76 = 32 Accept 31 or 31.76		1		1	1	1
			III.	Close to 0.75 for many throws **or** mean close to 0.75 or 32 is close to 35 or fits quite well with $(0.75)^n$ (1) Some further out e.g. 0.90 (1) Random process [these results are to be expected] (1)			3	3		3
				Question 6 total	**1**	**9**	**3**	**13**	**7**	**7**